

,Princeton University
Mathematical Intermediate Microeconomics (ECO 310)
Midterm Spring 2024

Instructions: You have 85 minutes to complete this exam, plus 10 minutes to upload your answers. There are 10 multiple choice questions and 2 short answer questions. **In the Multiple Choice section, simply indicate the correct answer. In the Short Answer section, show your work and explain your answers for full credit!**

- The exam is open-note, and calculators are permitted. However, please work **alone**.
- **If you are unsure whether you're interpreting a question correctly, state your interpretation and proceed (or email me if you prefer).**

The honor code applies, as usual.

Q1: Multiple Choice (40 points, 4 each)

CDDAD A D BCD B C

1. Jake cares about three attributes of cars: price, quality, and color. When comparing two cars, he prefers whichever is better in two attributes. Jake is:
 - (a) Rational
 - (b) Irrational because these preferences are incomplete
 - (c) Irrational because these preferences are intransitive
 - (d) Two of these are correct

Answer: C. They are complete—for any two cars, one of them has to be best in two attributes, so this one is preferred. But they are intransitive (there was a voting example like this in precept 1). For example, suppose comparisons are that in terms of price, car 1 beats car 2 beats car 3; in terms of quality, car 2 beats car 3 beats car 1; and in terms of color, car 3 beats car 1 beats car 2. Then car 1 is preferred to car 2 (better for price and color), car 2 is preferred to car 3 (better in terms of price and quality), but car 1 is NOT preferred to car 3, as would be required by transitivity, since car 3 beats car 1 in terms of both quality and color. if car 1 beats car 2 because of price a ; and this leaves at most one attribute where the other can win.

2. In which of the following situations *might* (depending on prices and income) it be optimal to spend your entire budget on good 1?
 - (a) If $u(x_1, x_2) = x_1 + x_2^2$
 - (b) If $u(x_1, x_2) = x_1^2 + x_2$
 - (c) If $u(x_1, x_2) = x_1^2 x_2$
 - (d) In two of these situations
 - (e) In three of these situations

Answer: D (A and B are both concave preferences so this could happen; it could not in C, where zero of either good means zero utility).

3. A consumer's preferences over good 1 and good 2 are described by $u(x_1, x_2) = x_1 x_2^3$. If this consumer behaves optimally, he will:
 - (a) Buy 3 times as much good 2 as good 1.
 - (b) Buy 4 times as much good 2 as good 1.
 - (c) Spend 4 times as much on good 2 as good 1.
 - (d) None of these is correct.

Answer: D. These are Cobb Douglas preferences, and the 3:1 exponent means they will SPEND 3 times as much on good 2 as good 1, i.e. $p_2 x_2 = 3 p_1 x_1$. (Specifically, they will spend $\frac{3}{4}$ income on good 2 and $\frac{1}{4}$ income on good 1).

4. A consumer initially buys 5 oranges and 10 apples. Then an apple tax increases the price per apple by \$0.50, but the consumer is at the same time given a \$5 fruit subsidy. Which of the following best describes this situation?

- (a) The consumer is most likely better off now (with the tax and subsidy) than he was originally.
- (b) The consumer is most likely worse off now (with the tax and subsidy) than he was originally.
- (c) The consumer is definitely indifferent between his original and current situations.

Answer: A. The subsidy lets them buy exactly their original bundle, since 10 apples have increased in cost by \$5. But most likely they can reoptimize at the new prices, and get something even better than the old bundle if they substitute some oranges for apples.

5. A consumer gets utility $u(L, R) = \min\{L, R\}$ from L left shoes and R right shoes. If $m = 10$, $p_L = 1$, and p_R increases from 1 to 4, the change in demand for right shoes due to the income effect is:

- (a) 0
- (b) -1
- (c) -2
- (d) -3
- (e) None of these is correct

Answer: D. With perfect complement preferences the consumer optimally buys $L = R = \frac{m}{p_L + p_R}$. So initially $\frac{10}{1+1} = 5$ right shoes and eventually $\frac{10}{1+4} = 2$ right shoes. But we know that with perfect complements, there is no substitution effect (no reoptimization: the best way to maintain utility is with the original bundle). So this 3-unit drop is entirely income effect.

6. A consumer expects to end up with final wealth $w_0 - 200$ if an accident occurs (this happens with probability p), and $w_0 + 100$ otherwise. They tell you that their expected utility from accepting this gamble is 100. Which of the following would let you conclude that the consumer is risk-averse?

- (a) If you knew that $u(w_0 + 100 - 300p) > 100$
- (b) If you knew that $u(w_0 + 100 - 300p) < 100$
- (c) If you knew that $u(w_0 - 200) < 100$
- (d) None of these is correct

Answer: A. They are risk averse if utility of expected wealth beats expected utility. Expected wealth here is $p(w_0 - 200) + (1 - p)(w_0 + 100) = w_0 + 100 - 300p$ and we're given that expected utility is 100.

7. In Lecture 10, we saw the following “paradox”: many people prefer saving 200 of 600 people for sure, to a risky option with a $\frac{1}{3}$ chance of saving all 600 people and a $\frac{2}{3}$ chance of saving nobody. But they prefer a risky option with a $\frac{1}{3}$ chance of killing nobody and a $\frac{2}{3}$ chance of killing everybody, to an option that kills 400 people for sure.

If people care about the # people alive in the end, then

- (a) This is inconsistent with EU maximization
- (b) This violates consequentialism

- (c) This violates independence
- (d) Two of these are true
- (e) All of these are true

Answer: D (A and B are both true. This describes identical choices in two different ways, eg describes 200 people being alive initially as 200 saved, and then as 400 killed. Thus it is inconsistent with EU maximization, and specifically it's consequentialism that is violated, which says that people should care only about the ultimate lottery over outcomes, not how it is described).

8. Which of the following would a risk-averse consumer **definitely** prefer to a 50-50 \$0 or \$200 gamble? Pick **ALL** that apply.
- (a) \$90 lump-sum
 - (b) \$110 lump-sum
 - (c) A 50-50 \$50 or \$150 gamble
 - (d) A 50-50 \$10 or \$190 gamble.

Answer: B,C,D are all correct. The gamble has expected wealth \$100, so B is clearly better (more money with less risk) but A may not be (less risk is good, less money is bad). C and D are also better as they are keeping expected wealth at \$100, but moving the outcomes closer together by equal amounts is reducing the risk, which a risk-averse consumer likes.

9. On Monday, Homer Simpson buys a bundle of food and drink that costs \$100. On Tuesday, he buys a different bundle, for \$80, which *would have cost \$100 on Monday*. In which of the following situations does Homer violate WARP?
- (a) If the Monday bundle would cost more than \$80 at Tuesday's prices.
 - (b) If the Monday bundle would cost less than \$80 at Tuesday's prices.
 - (c) If the Monday bundle would cost less than \$100 at Tuesday's prices
 - (d) Homer does not violate WARP
 - (e) Not enough information given.

Answer: B. On Monday, the bundle purchased costs \$100, and the Tues bundle would also have cost \$100; this reveals that the Monday bundle is weakly preferred. So Homer violates WARP if his Tues choice suggests a strict preference for the Tues bundle, namely if the Monday bundle would cost less than \$80 at Tuesday's prices.

10. Jessica has perfect substitute preferences over pens and pencils: she wants as many things to write with as possible, but doesn't care whether pens or pencils. She sets up a Lagrangian to find her best affordable bundle. Which of the following best describes which non-negativity constraints will bind, assuming **pens are cheaper**?
- (a) Non-negativity constraints bind for both pens and pencils
 - (b) Non-negativity constraints are NON-binding for both pens and pencils.
 - (c) Non-negativity constraints bind for pencils but not pens
 - (d) Non-negativity constraints bind for pens but not pencils

(e) Not enough information given to answer this question

Answer: C. If pens are cheaper, she buys pens only but no pencils. This indicates that non-negativity constraints bind for pencils but not pens. (She wants a positive pen quantity anyway; but if nonnegativity wasn't a rule, she'd buy a negative # pricey pencils and use the money to buy lots of cheap pencils).

Short Answer

Q2: Expected Utility

Consider a decision-maker with utility from wealth $u(w) = w^{1-\alpha}$.

- a. (10) For a gamble that offers equal chances of either \$0 or \$200, write out the equation describing the certainty equivalent (assuming no initial wealth for this part only). If α increases, does the certainty equivalent increase or decrease?

Answer: it is given by

$$\begin{aligned} u(CE) &= EU \\ CE^{1-\alpha} &= \frac{1}{2}0^{1-\alpha} + \frac{1}{2}(200^{1-\alpha}) \Rightarrow CE = \left(\frac{1}{2}\right)^{\frac{1}{1-\alpha}} \cdot 200 \end{aligned}$$

As α increases, this person becomes more risk averse and so their certainty equivalent decreases (this is the intuition, they'd take less in order to get a sure thing). Specifically, as α rises, $1 - \alpha$ falls so $\frac{1}{1-\alpha}$ rises so $2^{\frac{1}{1-\alpha}}$ rises so $\left(\frac{1}{2}\right)^{\frac{1}{1-\alpha}}$ falls.

- b. Now suppose the DM can invest as much of their initial wealth w_0 as they like in a double-or-nothing asset: their investment doubles in value with chance p , but if this doesn't happen, they lose everything they invested. (For example, \$100 either grows to \$200 or vanishes).

- i. (8) Write out the expected utility from investing \$x.

Answer: the double-or-nothing investment is like an interest rate r that equals either +1 with chance p , or -1 with chance $1 - p$. Final wealth if \$x is invested at rate r is $w_0 + rx$. So, for this person, final wealth is $w_0 + x$ with the chance p that they double their money, and $w_0 - x$ with the chance $1 - p$ that they lose their investment. Thus,

$$\begin{aligned} EU &= pu(w_0 + x) + (1 - p)u(w_0 - x) \\ &= p(w_0 + x)^{1-\alpha} + (1 - p)(w_0 - x)^{1-\alpha} \end{aligned}$$

- ii. (12) Find (or characterize) the optimal investment \$x.

Answer: the interior optimal investment that maximizes EU in (a) solves $EU'(x) = 0$. Here,

$$p(1 - \alpha)(w_0 + x)^{-\alpha} - (1 - p)(1 - \alpha)(w_0 - x)^{-\alpha}$$

So

$$\begin{aligned} EU'(0) &\Leftrightarrow p(w_0 + x)^{-\alpha} = (1 - p)(w_0 - x)^{-\alpha} \\ &\Leftrightarrow \frac{p}{(w_0 + x)^\alpha} = \frac{1 - p}{(w_0 - x)^\alpha} \\ &\Leftrightarrow \frac{p}{1 - p} = \left(\frac{w_0 + x}{w_0 - x}\right)^\alpha \\ &\Leftrightarrow \left(\frac{p}{1 - p}\right)^{\frac{1}{\alpha}} = \frac{w_0 + x}{w_0 - x} \\ &\Leftrightarrow x = \frac{\left(\frac{p}{1 - p}\right)^{\frac{1}{\alpha}} - 1}{\left(\frac{p}{1 - p}\right)^{\frac{1}{\alpha}} + 1} w_0 \end{aligned}$$

Now, recall that the FOC is only valid if the consumer is risk averse, which happens here if $\alpha > 0$ (this is a CRRA utility function): if $\alpha \leq 0$, then this consumer will invest either all or none of their wealth, depending on p . So let us assume $\alpha > 0$. In this case, the above FOC is valid provided that gives a nonnegative value for x , namely provided that $p \geq \frac{1}{2}$. If $p > \frac{1}{2}$, they invest an amount that's strictly positive and strictly below w_0 . If $p < \frac{1}{2}$, their best option given nonnegativity constraints is to invest zero. Also notice that the optimal investment increases in p , and decreases in α .

Grading note: this last part was worth generally 2.5 points, so if I took off 2.5 and said "characterize/check corners" that's what I meant.

Q3: Demand

Melsi (my puppy) has preferences over cookie consumption today (C_1) and cookie consumption tomorrow (C_2) given by $u(C_1, C_2) = v(C_1) + v(C_2)$, where $v : \mathbb{R} \rightarrow \mathbb{R}$ is an *increasing and concave* function (eg log or square root).

She has w_1 cookies today. She expects any cookies that she saves to magically double overnight, but otherwise does not expect to receive any additional cookies tomorrow.

- a. (5) Write out her budget constraint. (If she eats C_1 cookies today, how many does she save, and then how many can she eat tomorrow?)

Answer: cookies tomorrow are double her savings $w_1 - C_1$ from today. So $C_2 \leq 2(w_1 - C_1)$. In standard form, $2C_1 + C_2 \leq 2w_1$, or $C_1 + \frac{1}{2}C_2 \leq w_1$

- b. (6) Are her preferences over bundles (C_1, C_2) convex? Explain.

Answer: yes. Here,

$$\frac{MU_1}{MU_2} = \frac{v'(C_1)}{v'(C_2)}$$

We are given that v is concave, so as $C_1 \uparrow$ and $C_2 \downarrow$, $v'(C_1)$ falls and $v'(C_2)$ rises. Both of these changes mean that $\frac{MU_1}{MU_2}$ falls, thus these pass our convexity test.

- c. (9) Describe, as precisely as you can, how to find her best affordable bundle (C_1, C_2) as a function of w_1 . (You can't solve for it exactly without knowing v , but write out the equations you would solve if you did know v . You may use any method).

Answer with the intuitive method: in standard form, her budget constraint is $2C_1 + C_2 \leq 2w_1$. So the "price" of C_1 is 2, and the "price" of C_2 is 1, and "income" is $2w_1$. She looks for the pair (C_1, C_2) solving two conditions:

$$\begin{aligned} \text{optimal spending} & : \quad \frac{v'(C_1)}{v'(C_2)} = \frac{p_1}{p_2} \Rightarrow \frac{v'(C_1)}{v'(C_2)} = \frac{2}{1} \\ \text{spend all money} & : \quad C_2 = 2(w - C_1) \end{aligned}$$

- d. (10) For this part only, specialize to $v(C) = \ln C$. What bundle (C_1, C_2) most efficiently i.e. cheaply attains utility u^* ?

Answer: to allocate \$ optimally between two goods, she sets

$$\frac{v'(C_1)}{v'(C_2)} = 2$$

(as in (c)). But with this utility function, the condition becomes

$$\frac{C_2}{C_1} = 2 \Rightarrow C_2 = 2C_1$$

Then the second condition is that, instead of trying to spend less than some set income, she needs to get utility u^* . So she needs:

$$\ln C_1 + \ln C_2 = u^*$$

Plugging in the optimal spending condition,

$$\begin{aligned} \ln(C_1) + \ln(2C_1) &= u^* \Rightarrow \ln(2C_1^2) = u^* \Rightarrow 2C_1^2 = e^{u^*} \\ \Rightarrow C_1^2 &= \frac{1}{2}e^{u^*} \\ \Rightarrow C_1 &= \sqrt{\frac{1}{2}}e^{\frac{1}{2}u^*} \end{aligned}$$

and thus

$$C_2 = 2C_1 = \sqrt{2}e^{\frac{1}{2}u^*}$$

(Note that nonnegativity isn't an issue here).

Grading note: I only gave 5/10 to the (unfortunately many) students who calculated the best affordable bundle instead (Marshallian demand), as that indicated a lack of understanding of the compensated demand concept.